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Hybrid Smart Insole for Diabetic Neuropathy with Health Monitoring and Cloud-Base Analytics

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Abstract: Diabetic neuropathy is a condition that causes nerve damage due to high blood sugar levels, leading to loss of sensation, especially in the feet. The smart insole system is designed for diabetic neuropathy patients who are at high risk of foot injuries, infections, and ulcers. It integrates multiple sensors, including a force sensor to track pressure distribution, a moisture sensor to detect excessive perspiration that may cause fungal infections, a temperature sensor to identify early signs of inflammation or infection, and a touch sensor to ensure proper foot placement detection. The system features a hybrid power solution with solar energy and a rechargeable battery for continuous and sustainable operation. Collected data is securely stored in the Ubidots cloud server for real-time remote monitoring and analysis by healthcare professionals. If the system detects excessive pressure, moisture, or temperature, it sends an SMS alert via a GSM module, notifying the patient or caregiver to take immediate action. This intelligent insole system provides real-time tracking and proactive alerts to prevent complications while ensuring energy efficiency, continuous monitoring, and early warning capabilities. Ultimately, it enhances foot health management and improves the quality of life for diabetic patients.

Keywords -Diabetic neuropathy, smart insole, DiabeTrack , foot, Ubidots, Plantar Pressure Analysis

I. INTRODUCTION

Diabetic neuropathy is a common and serious complication of diabetes that results from prolonged high blood sugar levels, leading to nerve damage, especially in the extremities. One of the most affected areas is the foot, where patients often experience a loss of sensation. This reduced sensitivity increases the risk of undetected injuries, infections, pressure ulcers, and even amputation in severe cases. Proper foot care and continuous monitoring are essential for diabetic patients, particularly those with neuropathy, to prevent complications and maintain a better quality of life. To address these challenges, this project presents a smart insole system designed specifically for diabetic neuropathy patients. The insole integrates a range of sensors and intelligent features to monitor foot pressure, temperature, moisture, and shoe usage in real time. The system is capable of detecting abnormal conditions and sending immediate alerts to both patients and healthcare providers. By combining wearable technology, sensor integration, and cloud-based analytics, the proposed solution aims to provide early warnings, ensure continuous foot health monitoring, and support personalized treatment interventions.

Machine learning and deep learning algorithms, specifically TabNet, analyze EHR data to detect diabetic autonomic neuropathy (DAN) in the heart and kidneys. A multimodal system incorporating thermal imaging, multi-spectral imaging, and 3D imaging monitors foot health. Thermal imaging detects temperature changes, multi-spectral imaging assesses blood oxygenation, and 3D imaging provides detailed foot surface mapping. This approach helps identify key biomarkers such as blood circulation, pre-ulcer formation, swelling, and toe health, aiding in the early detection and monitoring of diabetic foot complications [1,2]. Four orthotic insoles, including a computer-designed model, were tested to reduce foot pressure in diabetic patients with peripheral neuropathy. The findings highlight the importance of personalized assessments and custom insoles for effectively managing plantar pressure in early-stage diabetes [3]. A non-invasive system utilizing a Raspberry Pi and DS18B20 temperature sensors detects diabetic peripheral neuropathy by monitoring foot temperature, a key indicator of neuropathy and ulcer risk. This affordable and user- friendly system delivers results within 20-25 minutes, with data analyzed using Python and displayed on an LCD screen. The study reveals that diabetic patients exhibit higher foot temperatures (35-40°C) compared to non-diabetics (30-34°C), enabling early detection and prevention of

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complications [4]. SMEAD is an IoT-based system using wearables and MEDIBOX for diabetes monitoring, insulin storage, and reminders, with Blockchain-secured real-time data for doctor monitoring [5].

II. LITERATURE REVIEW

The Dia shoe is a mobile-based device designed to monitor foot temperature, humidity, weight, and step count in diabetic patients. It syncs with a mobile app for real-time tracking and alerts, helping prevent ulcers and enabling early intervention [18]. Its electronic system includes pressure, temperature, and humidity sensors placed at key foot areas, with data stored in a battery-powered device for analysis. This system aids in detecting foot issues and could provide real-time feedback to prevent skin breakdown [6].An IoT-based real-time monitoring system tracks foot pressure and temperature in diabetic patients. Since diabetic neuropathy reduces sensation, leading to injuries, ulcers, and amputations, this system uses sensors and cloud connectivity to provide real-time data and SOS alerts to patients and healthcare centers. It enhances early detection and prevention of complications, with future improvements focusing on better sensor accuracy and features [7]. An IoT-based system monitors diabetic ketoacidosis (DKA) in type 1 diabetes by measuring breath acetone levels using a FIGARO TGS 822 gas sensor. An Arduino with an ESP8266 module transmits real-time data for remote monitoring. A diabetic foot ulcer device utilizes temperature and pressure sensors for early detection, while a vibration motor and accelerometer aid in healing and progress tracking. These advanced, non-invasive technologies offer cost-effective solutions for improved diabetes management [8,9].

DiabeTrack is a smart footwear system that monitors plantar pressure in diabetics, preventing foot ulcers. It uses flexible sensors, Bluetooth connectivity, and machine learning, with the Random Forest model achieving 94.7% accuracy. The system enables early detection and proactive care, improving patient outcomes. Future enhancements will refine predictive accuracy [10]. DiabeTrack is a wireless wearable system that helps diabetic patients monitor foot ulcers and track daily activity. It offers an affordable solution for managing foot health, overcoming challenges related to movement and sensory neuropathy in daily life [11]. A real-time IoT system enables remote monitoring of diabetic patients by transmitting glucose data. Using machine learning, it achieves 97.87% accuracy in classification, enhancing diabetes management and early intervention. A machine learning model utilizing CNN, SVM, and Random Forest analyzes 24-hour HRV data from Holter ECG to detect Cardiovascular Autonomic Neuropathy (CAN) in diabetics with 98.3% accuracy, providing a non-invasive solution for early diagnosis and prevention of cardiac complications.[19].

A machine learning model incorporating CNN, SVM, and Random Forest analyzes 24-hour HRV data from Holter ECG to identify Cardiovascular Autonomic Neuropathy (CAN) in diabetics with 98.3% accuracy, offering a non-invasive method for early detection and cardiac complication prevention [20]. The technology used includes a standard optical scanner for capturing pressure distribution and color-coded mapping to highlight high-risk ulceration areas. This method leverages tissue blanching analysis to provide an effective and low- cost screening tool for early diabetic foot detection [12]. The smart insole, crafted from natural rubber (Hevea brasiliensis), is designed for diabetic foot patients and integrates a pressure monitoring system with an electronic circuit to track plantar pressure. It also includes a laser-based signal irradiating cell to support tissue regeneration and healing. This innovative technology helps monitor foot pressure while aiding in the prevention and treatment of diabetic foot complications [13].

Wearable pressure sensors, cloud technology, and artificial neural networks (ANNs) monitor foot pressure and detect ulcer risks. ANNs analyze data, while cloud connectivity enables remote monitoring and intervention by healthcare providers [14].The Support Vector Machine (SVM) classifier identifies thermal biomarkers for diabetic peripheral neuropathy (DPN) by analyzing temperature recovery after cold provocation, aiding in early detection and assessment[15]. Machine learning detects Diabetic Peripheral Neuropathy (DPN) through plantar pressure analysis using Tekscan F-Scan sensors. SVM and MLP models classify DPN with over 92% accuracy, enabling non-invasive early detection. Future improvements focus on severity classification and feature selection [16]. A non-invasive microwave sensor with CSRRs detects foot tissue changes, distinguishing normal and ulcer-affected feet. This portable tool enables early detection, supporting timely intervention and improved diabetic foot care [17].

III. PROPOSED SYSTEM

The proposed smart insole system is designed to continuously monitor the foot conditions of diabetic neuropathy patients using a combination of sensors, a microcontroller, and wireless communication.

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From figure 3.1 and 3.2 shows three waterproof flex sensors are strategically placed under the big toe, little toe, and heel to detect pressure distribution and changes in posture. A DS18B20 digital temperature sensor, which operates via the 1-Wire protocol and includes a pull-up resistor for stable data communication, is embedded in the heel to measure localized foot temperature and detect early signs of inflammation or infection. A moisture sensor is also installed in the heel region to detect excessive perspiration, which may lead to fungal infections. Because the moisture sensor typically outputs a low- voltage signal, an operational amplifier is used to boost the signal strength for accurate detection by the microcontroller. To make the analog signals from these sensors compatible with the STM32F103C8T6 microcontroller's ADC input range, three voltage divider circuits are used. These ensure proper voltage scaling, prevent signal distortion, and protect the microcontroller from overvoltage. A capacitive touch sensor and an inductor coil are integrated into the insole to determine whether the shoe is being worn.

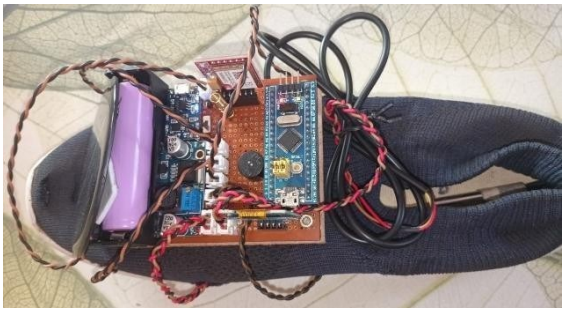

Figure 3.1: Smart Insole System Prototype-1

Figure 3.2: Smart Insole System Prototype-2

The STM32F103C8T6, an ARM Cortex-M3-based microcontroller, serves as the processing unit for collecting, analyzing, and transmitting data. For communication, a SIM800L GSM module is employed to send alerts and updates to the patient's smartphone or computer in real time. The entire system is powered by a hybrid power supply consisting of a 3.7V/2200mAh Li-Ion battery, managed through a TP4056 charging module, and supported by solar panels for extended battery life. A toggle switch allows the user to manually control the device's power state. All collected data, including pressure points, temperature, moisture levels, and shoe-wearing status, is uploaded to the Ubidots cloud platform for secure storage and real- time remote monitoring by healthcare professionals. This enables early diagnosis, preventive care, and personalized treatment planning, ultimately enhancing the management of diabetic foot complications.

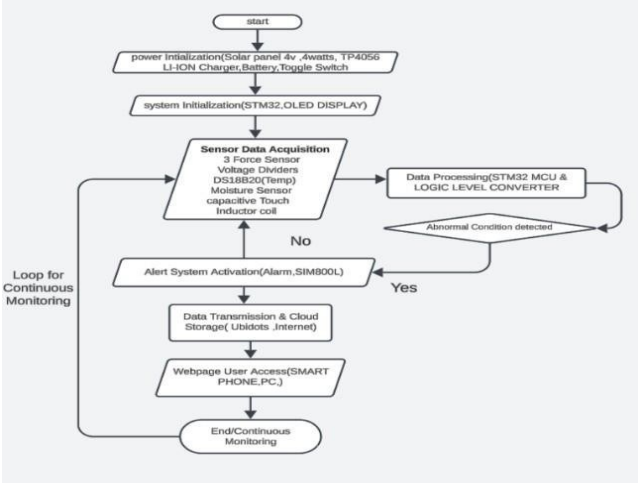

Figure 3.3 system Flowchart for Continuous Monitoring and Alert System

From Figure 3.3 shows the smart insole shoe monitors diabetic neuropathy patients by detecting posture, temperature, and moisture levels using force sensors, a DS18B20 sensor, and a moisture sensor with an amplifier. A capacitive touch sensor and inductor coil verify if the shoe is worn. The STM32F103C8T8 microcontroller processes sensor data, with voltage dividers adjusting signals and a logic level converter ensuring stable communication. If abnormalities are detected, an alarm is triggered, and alerts are sent via the SIM800L module to connected devices. Data is stored in the Ubidots cloud for remote monitoring, enabling real-time health tracking and early intervention.

IV. Smart Insole Data Visualization and Remote Monitoring via Ubidots

Figure 4.1: Smart Shoe System Overview with Ubidots Integration [21]

Figure 4.1 illustrates the overall concept of the smart shoe designed for diabetic neuropathy patients. It shows a patient wearing the smart insole system, which collects vital foot health data including pressure, temperature, and moisture. The data is transmitted to

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the cloud via GSM for remote monitoring. Healthcare professionals and caregivers can access the real-time data through Ubidots, enabling proactive care and early intervention. The system is powered by a hybrid energy source, supporting both solar and battery charging for uninterrupted monitoring throughout the day and night.

Figure 4.2: Big Toe Pressure Monitoring via Ubidots Dashboard

Figure 4.2 displays the real-time pressure reading at the big toe region of the smart insole, visualized through the Ubidots dashboard. Monitoring such pressure levels is crucial in detecting abnormal foot stress, preventing ulcers, and ensuring early intervention for diabetic neuropathy patients.

Figure 4.3: Little Toe Pressure Monitoring via Ubidots Dashboard

Figure 4.3 shows the live pressure reading detected at the little toe region of the smart insole system using the Ubidots interface. Continuous tracking of such localized pressure helps in identifying abnormal stress points, reducing the risk of pressure sores and aiding in the effective management of diabetic foot health.

Figure 4.4: Heel Pressure Monitoring via Ubidots Dashboard

Figure 4.4 illustrates the real-time pressure value at the heel region of the foot as captured by the smart insole system. Monitoring heel pressure is essential for identifying prolonged stress areas, which can lead to ulcers or injuries, especially in diabetic neuropathy patients. This data aids in early detection and preventive care.

Figure 4.5: Moisture Level Monitoring via Ubidots Platform

Figure 4.5 shows the moisture level inside the smart insole, as displayed on the Ubidots dashboard. Monitoring foot moisture is vital for diabetic patients, as excess moisture can lead to fungal infections and skin breakdown. The system provides real-time alerts if moisture levels exceed safe thresholds, enabling timely interventions and improved foot care.

Figure 4.6: Temperature Monitoring Display via Ubidots Platform

Figure 4.6 shows the real-time foot temperature data collected by the smart shoe and visualized through the Ubidots dashboard.

V. RESULT AND DICUSSION

The existing system detects temperature and pressure, storing abnormal data in the cloud but lacks a hybrid power source and does not monitor foot moisture, which is vital for infection prevention. The proposed smart insole integrates temperature, pressure, and moisture sensors, powered by a hybrid Li-ion battery and solar panel, with wear detection to ensure accurate data collection. It sends real-time alerts via the SIM800L GSM module and logs data on the Ubidots cloud. To handle intermittent GSM connectivity, data and alerts are stored locally and uploaded once the connection is restored, preventing information loss. Currently, the system uses fixed thresholds for alerts but does not adapt to individual

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foot biomechanics. Future plans include machine learning for personalized baselines and dynamic thresholds to improve accuracy and reduce false alarm. From Table 5.1 Shows compares the proposed smart insole system with the existing method across key performance metrics.

Feature	Existing	Proposed System
Sensors	Temp, Pressure	Temp, Pressure, Moisture
Power	Single source	Hybrid (Battery + Solar)
Wear Detection	No	Yes
GSM Handling	No backup storage	Local backup storage
Cloud Use	Limited logging	Real-time + continuous logging

Table 5.1: compares the proposed smart insole system with the existing method across key performance metrics.

Notably, temperature sensor accuracy improves from ±1.5°C to ±0.5°C, pressure accuracy from ±5% to ±2%, and moisture sensing is introduced with ±3% accuracy. Runtime extends from 6 hours to over 24 hours, and GSM data transmission success increases from 85% to 99% with local buffering. These advancements demonstrate the system's improved reliability, performance, and potential for early infection prevention in diabetic foot monitoring.

VI. CONCLUSION

In conclusion, the smart insole shoe is a cutting-edge solution for diabetic neuropathy management, offering continuous monitoring of posture, temperature, and moisture levels to prevent infections and complications. By integrating advanced sensors, real-time alerts, and cloud-based data storage, it enables proactive healthcare intervention, remote monitoring, and data-driven decision-making for personalized treatment. The hybrid-battery system ensures long-term usability, while its seamless connectivity enhances accessibility for both patients and healthcare providers. Future advancements, including AI-based predictive analytics and improved connectivity, will further enhance its capabilities, making it an essential tool in modern diabetic foot care and overall patient well-being. Future work includes AI- based predictive analytics, pressure mapping for gait analysis, improved battery efficiency, and expanded connectivity via 5G or LoRaWAN, with machine learning integration for adaptive feedback and personalized treatment.

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